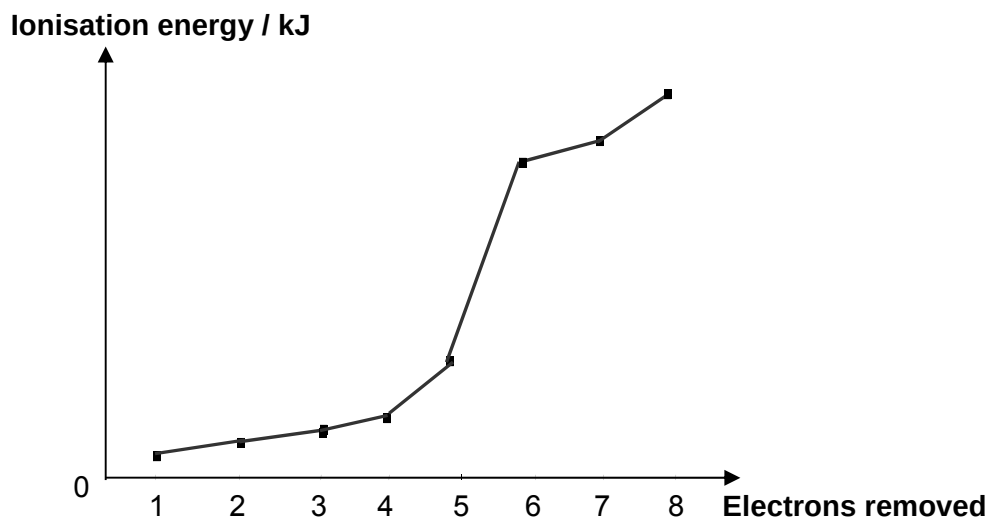


Section A (40 marks)

Answer **all** the questions in this section in the space provided.

- 1 (a) The first eight ionisation energies of element X are shown in the graph below.



- (i) To which Group does element X belong to? Explain your answer.

X: _____

W, X and Y are consecutive elements of increasing proton number in Periods 3. These elements form ions which are *isoelectronic* with element Ar.

- (ii) Explain the term 'isoelectronic'.

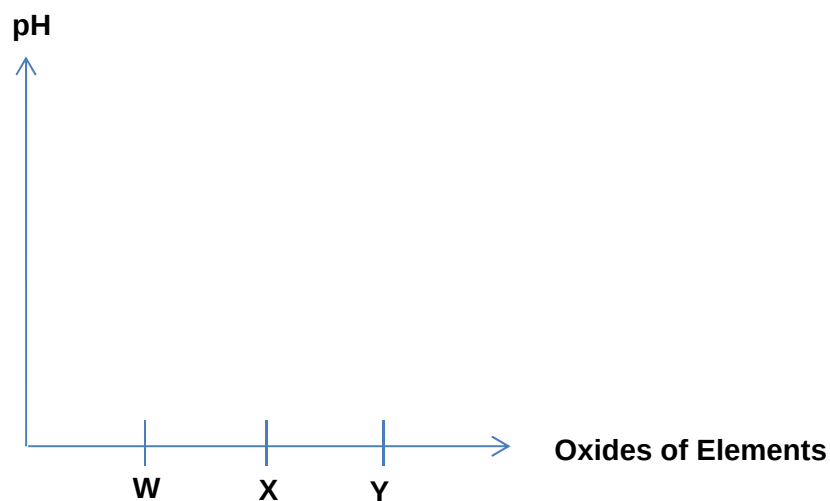
- (iii) Write the formula of the ions of W, X and Y which are isoelectronic with element Ar.

W: _____ X: _____ Y: _____

[5]

[Turn Over]

- 1 (b) (i) Sketch the pH trend of the resultant solutions when the oxides of elements W, X and Y are added to water in the axis below.



- (ii) Account for the trend above.

[4]

[Total: 9]

- 2 Nitrogen dioxide, NO_2 , dimerises to form dinitrogen tetroxide, N_2O_4 , which is the more stable form in the following equilibrium.



- (a) (i) Draw the dot-and-cross diagram of NO_2 and N_2O_4 .

NO_2	N_2O_4

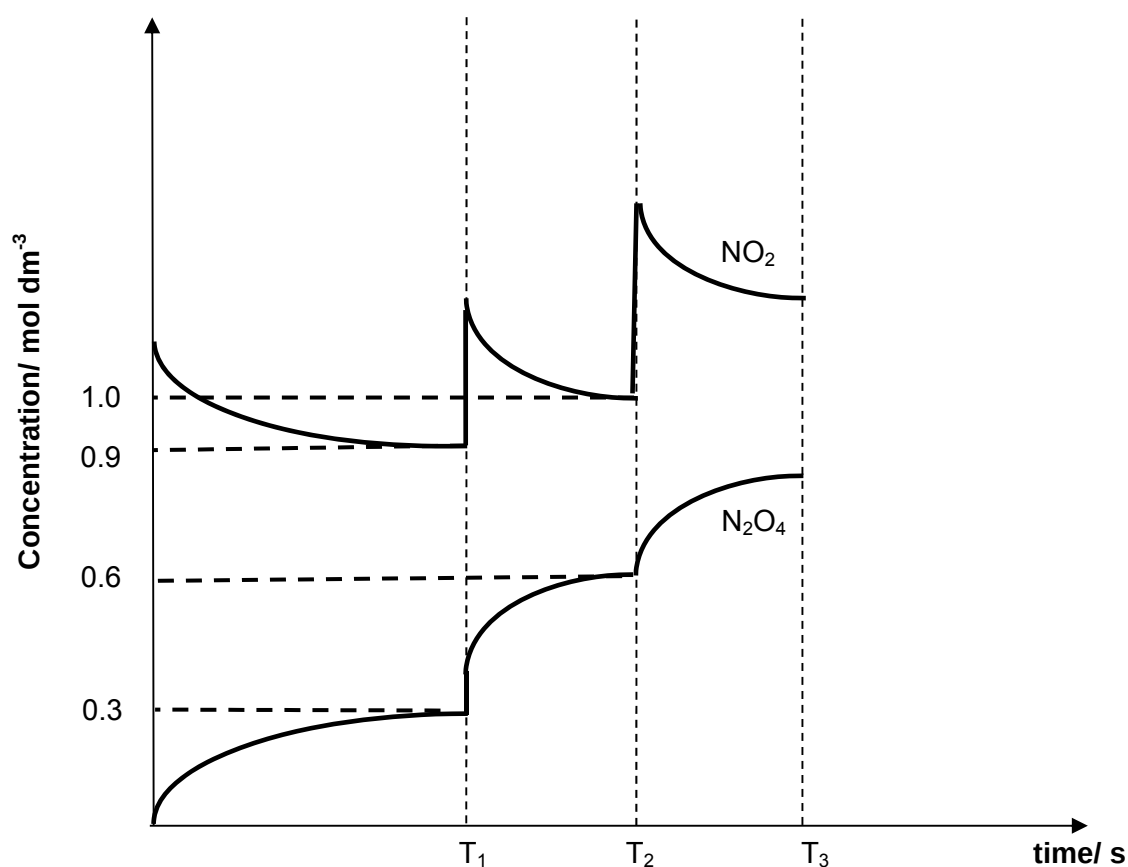
[Turn Over]

- 2 (a) (ii) Suggest a reason why the enthalpy change for the forward reaction is exothermic.

[3]

- (b) During an experiment, changes were made to the conditions in the reaction vessel. At each time, there was only one change made to the condition in the reaction vessel.

The change in the concentrations in the equilibrium mixture with time is shown in the graph below.



- (i) Write the K_c expression for the equilibrium.

[Turn Over]

- 2 (b) (ii) Calculate the K_c at T_1 , stating the units.

[3]

- (c) (i) What changes in condition were made at time T_1 and T_2 ?

T_1 : _____

T_2 : _____

- (ii) Sketch and label clearly on the given graph on page 4, the changes in the concentration of NO_2 and N_2O_4 , when the mixture was cooled at time T_3 . Explain your answer.

[5]

- (d) Sketch, on the axes below, how the **rates** of the forward and reverse reactions change from the time NO_2 was placed in the vessel to the time equilibrium was reached at time T_1 . Label your graphs clearly.



[1]

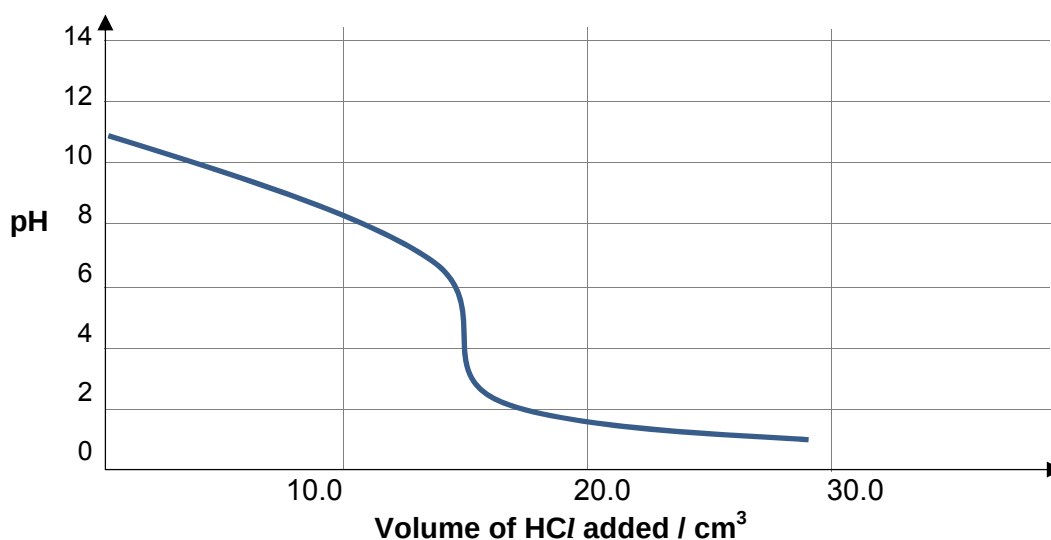
[Total: 12]

[Turn Over]

3 (a) Hydrazine, N_2H_4 , is a flammable liquid with an ammonia like odour. It has basic chemical properties like ammonia.

- (i) Suggest the equation for the reaction between hydrazine and water, assuming N_2H_4 is a monoprotic base.

The graph below shows the pH changes when 10.0 cm^3 of hydrazine was titrated with 0.05 mol dm^{-3} hydrochloric acid.



Indicator	pK_a	acid form	base form
Congo red	3 - 5	blue	red
Phenol Red	6.8 - 8.4	yellow	red
Thymolphthalein	9.3-10.5	colourless	blue

- (ii) From the table above, suggest a suitable indicator for the above titration.

Indicator: _____

- (iii) Suggest the colour change observed at the end-point.

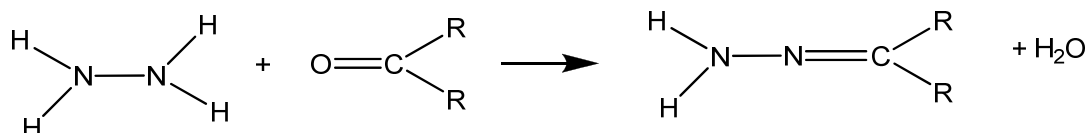
_____ to _____

[Turn Over

- 3 (a) (iv) When 7.5 cm^3 of HCl is added to excess hydrazine, the resultant solution act as a buffer. Write an equation to explain what happens when some solid sodium hydroxide is added to this buffer solution.

[4]

- (b) Hydrazine and its derivatives can undergo condensation reactions with carbonyl compounds as shown.



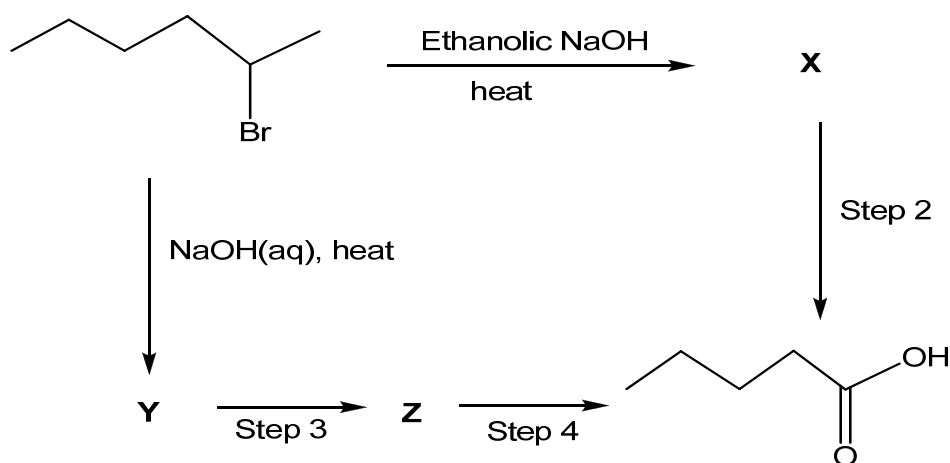
A similar condensation reaction takes place between $\text{CH}_3\text{CH}_2\text{NH}_2$ and propanone, CH_3COCH_3 .

Draw the structural formula of the product formed for the above reaction.

[1]

[Total: 5]

- 4 (a) Pentanoic acid could be synthesized from 2-bromo-hexane via two different synthetic routes as shown below, where X, Y and Z are different compounds.



[Turn Over]

- 4 (a) (i) Suggest the reagents and conditions for **steps 2 – 4**.

Step 2: _____

Step 3: _____

Step 4: _____

- (ii) State the type of reaction undergone in **step 3**.

Type of reaction: _____

- (iii) Draw the full structural formulae of Compounds **X**, **Y** and **Z**.

X	Y
Z	

[Turn Over

- 4 (a) (iv) Suggest a simple chemical test which you will carry out to distinguish between X and Y.

Reagents and Conditions: _____

Observations: _____

[9]

- (b) The following table shows the pK_a values of pentanoic acid, 3-hydroxypentanoic acid and pentanol.

Compound	pK_a
pentanoic acid	4.87
3-hydroxypentanoic acid	3.89
pentanol	16

- (i) Suggest a reason why the pK_a value of pentanoic acid is less than that of pentanol.

- (ii) Suggest a reason why 3-hydroxypentanoic acid is more acidic than pentanoic acid.

[4]

[Turn Over

- 4 (c) When concentrated sulfuric acid was added to 3-hydroxypentanoic acid, the mixture was refluxed. Only one organic product was obtained.
Draw the structural formula of the product formed.

[1]

[Total: 14]

[Turn Over]